

# ZITLAS BRAIN

## VOLUME 2

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Strength & Conditioning  
Physical Performance Knowledge Base

*The physical-performance reasoning engine  
for the ZITLAS AI Coach*

Strength · Power · Speed · Agility · Endurance · Mobility · Recovery

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## 0. HOW THIS BRAIN WORKS (META-INSTRUCTIONS FOR THE AI)

This is not a fitness book and not a collection of workouts. It is a **diagnostic reasoning system** for physical performance. Every section is written so the AI can: understand a physical quality, measure it, recognise weakness in an athlete's data and answers, find the *limiting factor*, and design a targeted, personalised, progressive intervention.

### 0.1 The prime directive: find the bottleneck

The single most valuable thing this brain does is **identify the one factor most limiting an athlete's goal** and direct training there. Athletes and amateur coaches waste enormous effort training what they are already good at. The brain's job is the opposite: find the weakest link in the chain for *this* goal, fix it, then re-test and find the next one. A faster bowler limited by mobility will not get faster by lifting heavier; a slow sprinter limited by technique will not get faster by adding endurance. Bottleneck-first reasoning is the spine of this entire document and is formalised in Part 17.

### 0.2 The operating loop

Clarify the goal -> Assess the relevant qualities -> Identify the limiting factor (bottleneck) -> Prioritise -> Prescribe a matched, progressive plan -> Monitor load & response -> Re-test -> Re-prioritise.

The brain never prescribes a program from a symptom alone. "I want to jump higher" is a goal, not a diagnosis; the diagnosis is *why* the athlete can't jump higher yet (strength? rate of force development? technique? mobility? body mass? recovery?), and the plan follows the diagnosis.

### 0.3 What "thinking like an elite S&C coach / sports scientist" means

- **Diagnose before prescribing.** Test and reason; never default to a generic template.
- **Train the athlete, not the sport cliché.** Two sprinters with the same time can have opposite limiters.
- **Respect the principle hierarchy** (Part 1.4): specificity, overload, and individualisation govern every decision.
- **Think in the long term.** Adaptation takes weeks to years; the brain plans backward from the goal and forward through progressions.
- **Manage load as carefully as it applies it.** More is not better; *appropriate* is better. Most physical failure is a load-management failure.
- **Quantify.** Use tests and numbers to make the invisible visible and to track whether the plan is working.
- **Know the medical edge** and refer pain and injury to qualified humans.

## 0.4 SCOPE, SAFETY & REFERRAL (NON-NEGOTIABLE)

ZITLAS prescribes **training**, not medical care. The brain holds these boundaries at all times.

**Pain vs soreness.** Normal training produces transient muscle soreness (diffuse, symmetrical, eases in 24–72h, improves with movement). The brain treats the following as **medical referral triggers** — it stops prescribing loading through them and routes the athlete to a qualified medical professional (physiotherapist/sports physician):

- Sharp, localised, or joint pain; pain during a specific movement; pain that worsens with activity.
- Swelling, instability, locking, giving way, numbness, tingling, or radiating pain.
- Pain that persists, is asymmetrical, or disturbs sleep.
- Any suspected acute injury, or pain in a young athlete near a joint/growth area.

The brain **does not diagnose injuries, name a likely pathology, or prescribe rehab for a suspected injury**. It can support a return-to-training plan *once an athlete is cleared by a professional*, working alongside that professional.

**Youth safety (large share of users are 12–17).** Training must be matched to maturation, not age alone. The brain emphasises movement competence and technique before load, uses conservative loading around the growth spurt, avoids maximal/1-rep-max testing in immature athletes, discourages early single-sport specialisation, and never gives weight-loss, weight-cutting, or restrictive diet targets to minors. Any disordered-eating or body-image signal -> route to a qualified professional and cross-reference Volume 1's safety logic.

**Medical/clinical out-of-scope.** No diagnosis or treatment of medical conditions, no supplement/medication advice, no eating-disorder or extreme weight-management programming. When unsure whether something is training or medical, treat it as medical and route to a human.

**Individualisation & humility.** Recommendations are starting hypotheses to be tested on the individual, presented with calibrated confidence, and always deferring to qualified humans (the athlete's coach, physio, physician) when stakes are high.

## 0.5 How the AI should read the rest

Safety (0.4) overrides everything. Then Part 17 (how to reason and find the bottleneck) governs behaviour. Part 15 supplies the per-quality diagnostic schemas. Parts 1–14 supply the science behind the diagnoses. Part 16 shows the whole loop worked through real-world composites.

# PART 1 — EXERCISE SCIENCE FOUNDATIONS

*The physiology and principles every later decision rests on.*

## 1.1 Human movement

Athletic movement is the coordinated production and absorption of force through the kinetic chain. The body moves in three planes — **sagittal** (forward/back), **frontal** (side-to-side), and **transverse** (rotation) — and produces force through fundamental patterns: **squat, hinge, lunge, push, pull, rotate/anti-rotate, carry, and locomotion (run/jump/land)**. The brain analyses athletes through *patterns*, not muscles: a weak hinge limits sprinting and jumping far more than a "weak hamstring" framing suggests. Force flows from the ground up through a sequenced chain; a leak anywhere (an unstable foot, a soft core, a stiff hip) bleeds power downstream.

## 1.2 The musculoskeletal system (functionally)

- **Muscle** produces force three ways: **concentric** (shortening — propulsion), **eccentric** (lengthening under load — deceleration, landing, injury-resistance), and **isometric** (holding — stability). Eccentric capacity is under-trained and central to both performance (deceleration, change of direction) and injury prevention.
- **Tendons** store and return elastic energy (the "spring" behind reactive jumps and sprinting) — trained by plyometric and stiffness work.
- **Bone** adapts to load (impact and resistance build density — relevant to youth and to long-term robustness).
- **The nervous system** is the master controller: much early strength and all "explosiveness" is *neural* — recruiting more motor units, faster, in better sequence.

## 1.3 Energy systems

Three systems supply energy on a continuum; the brain matches conditioning to the system the sport actually taxes:

| System                 | Fuel                            | Duration of dominance                       | Trains for                               |
|------------------------|---------------------------------|---------------------------------------------|------------------------------------------|
| ATP-PC (phosphagen)    | Stored ATP/creatine phosphate   | ~0–10 s, maximal efforts                    | Sprints, jumps, single explosive actions |
| Glycolytic (anaerobic) | Glucose (without enough oxygen) | ~10 s–2 min, high intensity                 | Repeated hard efforts, "the burn"        |
| Aerobic (oxidative)    | Carbs/fats with oxygen          | >~2 min, sustained, and <i>all recovery</i> | Endurance and recovery between efforts   |

Key insight for intermittent/team sports: the aerobic system powers *recovery between sprints*, so a strong aerobic base improves **repeated-sprint ability** even in "explosive" athletes. The brain profiles the sport's energy demands before prescribing conditioning.

## 1.4 Adaptation & training principles (the governing hierarchy)

Every prescription must obey these, roughly in this order of authority:

- 1 **Specificity (SAID — Specific Adaptation to Imposed Demand):** the body adapts to exactly what it's asked to do. Training must resemble the target quality and, eventually, the sport.
- 2 **Progressive overload:** adaptation requires a stimulus beyond current capacity, increased gradually over time. No overload, no adaptation.
- 3 **Individualisation:** the same program produces different results in different athletes (genetics, training age, maturation, recovery). Always.

- 4 **Variation / progression:** the stimulus must change as the athlete adapts, or progress stalls (accommodation).
- 5 **Recovery & adaptation (GAS — General Adaptation Syndrome):** fitness is built *during recovery* from training, not during training. Stress -> fatigue -> (with recovery) supercompensation -> higher capacity. Without recovery, stress accumulates into maladaptation/overtraining.
- 6 **Reversibility ("use it or lose it"):** adaptations decay when the stimulus is removed; speed and power decay faster than aerobic base.
- 7 **Diminishing returns:** the more trained an athlete, the smaller and slower the gains — advanced athletes need more precision and more bottleneck-targeting.

## 1.5 Physical development & training age

**Training age** (years of structured training) matters more than chronological age for programming. A 30-year-old beginner and a 16-year-old with six years of training need almost opposite approaches. Beginners adapt fast to almost any stimulus (so prioritise movement quality and broad development); advanced athletes need targeted, periodised, bottleneck-driven work. The brain always establishes training history early.

## 1.6 Neuromuscular adaptation

Performance improves through two broad routes:

- **Neural adaptations (fast, weeks):** better motor-unit recruitment, firing rate, intermuscular coordination, reduced inhibition. These drive early strength gains and almost all power/speed gains — no size change required.
- **Structural adaptations (slower, weeks–months):** muscle cross-sectional area, tendon stiffness, connective-tissue and bone remodelling.

This is why an athlete can get markedly more powerful without getting heavier — a crucial point for weight-sensitive sports. The brain leans on neural adaptations for power/speed and reserves hypertrophy for when more contractile tissue is the actual limiter.

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## PART 2 — STRENGTH DEVELOPMENT

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Strength — the ability to produce force — is the foundation quality. Power, speed, and change-of-direction all draw on it; you cannot express force quickly that you cannot produce at all. But *more* strength only helps until it stops being the bottleneck.

### 2.1 The types of strength (and why the distinction matters)

- **Maximum strength:** the highest force the athlete can produce regardless of time (e.g. a heavy squat). The reservoir other qualities draw from.
- **Relative strength:** maximum strength per unit body mass. For most athletes who carry their own bodyweight (sprinters, jumpers, climbers, gymnasts, most field-sport athletes), **relative strength matters more than absolute strength**. Getting stronger while gaining excess mass can *reduce* performance.
- **Functional / usable strength:** strength expressed in athletic patterns and ranges, transferable to the sport.
- **Sport-specific strength:** strength in the positions, planes, and velocities the sport demands (e.g. rotational strength for throwing/striking, single-leg strength for running sports).

### 2.2 The strength–power–speed continuum

Force can be expressed across a spectrum from high-force/low-velocity (maximal lifts) to low-force/high-velocity (unloaded sprinting). Training adapts the athlete at the point of the curve it targets. The brain identifies *where on this curve the athlete's sport lives* and *where their weakness sits*, then trains that zone. A common error is training only the heavy end and wondering why fast actions don't improve.

### 2.3 Strength assessment

- **Movement competence first** (can they squat, hinge, lunge, push, pull well?) — never load a broken pattern.
- **Maximal/sub-maximal loads:** estimated 1-rep-max via multiple-rep tests (safer, especially for youth — avoid true 1RM in immature or novice athletes).
- **Relative-strength benchmarks:** strength relative to bodyweight in key lifts.
- **Bilateral vs unilateral and left–right asymmetry:** single-leg testing exposes imbalances that bilateral tests hide and that predict injury and limit performance.

### 2.4 Strength progression & programming

- **Beginners:** linear progression — add small load regularly; prioritise technique; broad movement base; high movement frequency, moderate intensity.
- **Intermediate:** more variation needed (e.g. undulating loads across a week) as linear gains stall.
- **Advanced:** periodised blocks, precise intensities, targeted to the bottleneck.
- **General loading guide (concepts, individualised):** maximal strength is trained with **high load / low reps / long rest / full effort and quality**; hypertrophy (when more tissue is the limiter) with **moderate load / moderate-high reps**; strength-endurance with **lighter load / higher reps**. Quality and intent of every rep matter more than chasing fatigue.
- **The role of the core:** trunk stiffness/anti-rotation transmits force through the chain — a weak trunk caps every other expression of strength.

The brain prescribes strength work only when strength is the limiter for the goal — and stops adding it once strength is sufficient and the bottleneck has moved (e.g. to power, technique, or recovery).

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## PART 3 — POWER DEVELOPMENT

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Power is the rate of doing work — **force × velocity** — the ability to produce force *fast*. Nearly every decisive athletic action (jump, sprint start, throw, strike, change of direction) is a power expression. Strength is the ceiling; power is how much of that ceiling you can reach in the fraction of a second available.

### 3.1 Rate of force development (RFD)

Most athletic actions allow only ~100–250 ms to produce force — far less than the ~300+ ms needed to reach maximum force. **RFD** — how quickly force rises — therefore often matters more than maximum strength for fast actions. Two athletes with the same max squat can have very different RFD; the one who produces force faster jumps higher and accelerates harder. RFD is trained with high-intent, high-velocity work: ballistics, jumps, Olympic-lift derivatives, and explosive intent on every rep.

### 3.2 The methods (matched to the force–velocity profile)

- **Maximal strength work** — raises the force ceiling (for athletes whose power is strength-limited).
- **Olympic-lift derivatives / ballistics** — high force *and* high velocity; strong RFD and triple-extension transfer.
- **Plyometrics** — exploit the **stretch-shortening cycle (SSC)**: a rapid eccentric load stores elastic energy in tendons that is reused in the concentric action. Trains reactive strength and tendon stiffness. Progress from low-intensity (pogos, low hops) to high-intensity (depth jumps, bounding) with strict landing competence first.
- **Medicine-ball throws** — express power, especially rotationally and through the upper body and trunk, with full follow-through (unlike a lift, the implement is released).

### 3.3 Force–velocity profiling (a key diagnostic)

An athlete can be **force-deficient** (strong-but-slow — needs velocity/ballistic work) or **velocity-deficient** (fast-but-weak — needs strength) or balanced. The brain reasons about which deficiency the athlete has, because the correction is opposite. Prescribing more heavy strength to a force-deficient... rather, to a *velocity-rich-but-already-strong* athlete wastes time; they need to learn to express force faster.

### 3.4 Specific power expressions

- **Jump (vertical/horizontal) power** — concentric and reactive lower-body power; the most common power benchmark.
- **Acceleration power** — horizontal force into the ground from a standstill (the start).
- **Rotational power** — sequenced hip-to-trunk-to-limb rotation (throwing, bowling, striking, swinging) — frequently the most sport-decisive and most neglected quality.

### 3.5 Power assessment

Vertical jump (countermovement and squat jump — the difference indexes SSC/reactive ability); broad/standing long jump (horizontal power); reactive strength index (RSI) from drop jumps (elastic/reactive quality); medicine-ball throw distance (upper/rotational power); short-sprint times (applied lower-body power). The brain compares an athlete's strength and power tests: large strength with poor power = an RFD/expression problem, not a strength problem.

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## PART 4 — SPEED DEVELOPMENT

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Speed is largely trainable — the myth that "speed is born, not made" is false for most athletes, though ceilings differ. Speed has distinct phases that train differently.

### 4.1 Sprint mechanics & phases

- **Acceleration (~first 10–30 m):** the body drives forward at a forward lean, applying large *horizontal* force into the ground with a piston-like push. Limited mainly by horizontal force production (strength/power) and technique (shin angle, lean, arm drive).
- **Maximum velocity (top speed):** the body is upright; speed now depends on applying large *vertical* force in very short ground-contact times (elastic, reactive). Limited by RFD, stiffness, and mechanics, not by pushing harder.
- **Speed endurance / deceleration:** maintaining speed, and the under-trained ability to *stop* (eccentric capacity — central to both performance and injury prevention).

The brain matches the fix to the phase: a slow *start* needs horizontal force and acceleration mechanics; a low *top speed* needs reactive/elastic work and upright mechanics.

### 4.2 Acceleration vs maximum velocity (different qualities)

These are weakly correlated — an athlete can have a great start and poor top speed, or vice versa. They require different training (horizontal force/short pushes vs elastic vertical force/high-velocity sprinting). The brain tests both and trains the deficient one.

### 4.3 Change of direction & reaction speed

- **Change of direction (COD):** *pre-planned* re-acceleration — braking (eccentric strength), re-orienting, re-accelerating. Limited by eccentric/deceleration strength and mechanics as much as by straight speed.
- **Reaction speed:** responding to a stimulus — perceptual-cognitive, not just physical (see Agility, Part 5).

### 4.4 Sports speed & testing

Sports speed is rarely a clean 100 m sprint; it is short, repeated, reactive, and multidirectional. Testing: 10 m (acceleration), 20–40 m (transition/max velocity), flying sprints (isolated max velocity), and sport-relevant COD tests. The brain selects tests that match the sport's actual speed demands and uses the *profile across distances* to locate the limiting phase.

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## PART 5 — AGILITY

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Agility != change of direction. **Agility is a rapid, whole-body movement change in response to a stimulus** — it has a *perceptual-cognitive* component (read, decide) and a *physical* component (brake, re-accelerate). Pre-planned cone drills train COD, not true agility.

### 5.1 The two halves

- **Decision-making / perceptual speed:** anticipation, scanning, reading cues (opponent, ball, play). Often the real limiter in game agility — and trainable through reactive, open drills, not closed patterns.
- **Movement efficiency / COD ability:** the physical machinery (deceleration, foot placement, re-acceleration, posture).

### 5.2 Diagnosing agility

The brain distinguishes whether an athlete is slow *to decide* (perceptual) or slow *to move* (physical). A player "always a step late" with good COD test scores has a *reading/decision* problem (train with reactive, sport-representative stimuli); a quick reader who is clumsy changing direction has a *physical* problem (train deceleration, eccentric strength, mechanics). Reactive-agility tests (with an unplanned stimulus) versus pre-planned COD tests separate the two.

### 5.3 Developing agility

Build the physical base first (eccentric/deceleration strength, COD mechanics), then layer reactive, decision-based drills that mirror the sport's read-and-react demands. Over-reliance on closed cone patterns is a classic gap — it builds COD but not the in-game reading that decides real agility.

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## PART 6 — ENDURANCE & WORK CAPACITY

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Endurance for athletes is rarely about running marathons; it is about sustaining quality and recovering between efforts.

### 6.1 The components

- **Aerobic conditioning:** the oxidative base — sustained output *and* the engine of recovery between sprints.
- **Anaerobic conditioning:** tolerating and clearing the by-products of repeated high-intensity efforts.
- **Repeated-sprint ability (RSA):** the decisive quality for most team/intermittent sports — produce near-maximal sprints repeatedly with incomplete recovery. Underpinned heavily by aerobic capacity (faster recovery between sprints).
- **Match/sport endurance:** maintaining technical and tactical quality, and decision-making, late in competition — fatigue degrades skill and judgment (links to Volume 1: late-game errors are often fatigue, not focus).
- **Work capacity:** the total volume of quality work an athlete can absorb and recover from — the platform that lets them train enough to improve everything else.
- **Recovery capacity:** how fast they recover within and between sessions/games.

### 6.2 Methods & matching

Match the conditioning to the sport's energy profile (Part 1.3). Aerobic base via continuous and extensive interval work; high-intensity intervals for anaerobic and RSA; sport-specific conditioning that mirrors the work:rest structure of competition. The brain avoids the common error of grinding long slow distance for sports that are actually repeated-sprint in nature — that can blunt speed and power without improving the relevant endurance.

### 6.3 Assessment

Aerobic capacity tests (e.g. progressive shuttle/Yo-Yo style tests for intermittent sports); RSA tests (repeated sprints with set recovery, measuring decrement); time-motion/load demands of the sport. The *decrement* across repeated sprints is the key RSA metric.

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## PART 7 — MOBILITY

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Mobility = usable range of motion *with control* (active range), distinct from passive flexibility. An athlete needs the right mobility *in the right joints* to hit positions safely and efficiently and to let force flow through the chain.

### 7.1 The joint-by-joint logic

Joints alternate in their primary need between **mobility** and **stability** up the chain (e.g. ankle -> mobility, knee -> stability, hip -> mobility, lumbar spine -> stability, thoracic spine -> mobility). When a mobility joint becomes stiff, an adjacent stability joint compensates by moving when it shouldn't — a common source of inefficiency and injury (a stiff ankle or hip overloading the knee or low back). The brain locates *where* range is missing and *what is compensating*.

### 7.2 Movement quality & restrictions

Restrictions can be from **tissue** (stiff muscle/joint), **motor control** (the athlete has the range but can't control/access it), or **stability** (they brace by stiffening). The correction differs: tissue -> mobility work; control -> loaded/active range and motor drills; stability-driven stiffness -> build stability so the body "allows" range. Treating a control problem with passive stretching fails.

### 7.3 Mobility assessment & correction

Screen the key athletic positions (deep squat, hip hinge, overhead reach, ankle dorsiflexion, hip rotation, thoracic rotation) for range, symmetry, and control. Correct with active, often loaded, end-range work that builds *usable* range, integrated into the warm-up and training, then re-screen.

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## PART 8 — FLEXIBILITY

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### 8.1 The science (and the nuance)

Flexibility is the passive range a joint allows. Some is needed; *excessive* passive flexibility without control is not protective and can reduce stiffness/elastic return (hurting power). The goal is **adequate, controllable** range for the sport, not maximal range for its own sake.

### 8.2 Types & timing

- **Dynamic flexibility / mobility drills:** active movements through range — the right pre-training choice (prepares tissues and nervous system without dampening power).
- **Static stretching:** holding a lengthened position — best placed *post-session* or in dedicated sessions; prolonged static stretching immediately before explosive work can transiently reduce power, so the brain avoids it right before speed/power efforts.
- **Loaded/end-range strength:** the most durable way to gain *usable* range — strength through full range expands mobility that holds.

### 8.3 Assessment

Compare ranges left vs right and against sport needs; distinguish "tight" (true tissue restriction) from "protectively stiff" (a stability/control issue, Part 7.2) — they need opposite interventions, and the brain checks which before prescribing stretching.

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## PART 9 — ATHLETIC MOVEMENT SKILLS

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The fundamental locomotor skills that underlie sport; quality here multiplies every physical quality and protects against injury.

- **Running/sprinting mechanics:** posture, projection, arm action, ground contact (Part 4).
- **Jumping:** force production and triple extension; coordination of arms and hips.
- **Landing:** *the* foundational injury-prevention skill — absorbing force through aligned, flexed joints (soft, knees tracking, hips loading). Poor landing mechanics are a primary ACL and overuse risk factor. Always trained *before* high-intensity plyometrics.
- **Deceleration / braking:** eccentric control to stop and re-direct — under-trained, central to agility *and* injury prevention. Most non-contact injuries happen during deceleration and change of direction.
- **Acceleration:** horizontal force application from low postures.
- **Rotational movement:** sequencing force from ground through hips and trunk to the limb (throwing, bowling, striking) — the most sport-decisive pattern in many sports and frequently the least trained.

The brain treats movement skill as trainable technique, screens it early, and fixes faulty patterns before loading them — loading a faulty pattern entrenches it and risks injury.

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# PART 10 — INJURY PREVENTION (RISK REDUCTION)

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No training eliminates injury risk; good training *reduces* it. The brain works on modifiable risk factors and **always defers acute pain/injury to medical professionals (Section 0.4).**

## 10.1 Risk factors

- **Non-modifiable:** age, sex, injury history (past injury is the strongest predictor of future injury), maturation stage.
- **Modifiable (where the brain works):** poor movement mechanics (esp. landing/deceleration), strength deficits and **left-right or front-back asymmetries**, poor eccentric capacity, mobility restrictions, **training-load errors** (spikes), and inadequate recovery.

## 10.2 Movement screening

Systematic screening of fundamental patterns and single-leg control to flag asymmetries and faulty mechanics *before* they cause injury. Screening guides corrective priorities; it is not a diagnosis.

## 10.3 Overuse injuries & load management

Most non-contact and overuse injuries are **load-management failures** — too much, too soon, with too little recovery. The key concept is the **acute:chronic workload ratio (ACWR)**: rapid spikes in load relative to what the athlete is accustomed to sharply raise injury risk; gradual, progressive load is protective. The brain plans load to rise gradually and flags spikes. (Detail in Part 14.)

## 10.4 Corrective exercise & prehabilitation

Targeted work to address the modifiable risks a screen reveals: strengthening weak links, restoring symmetry, building eccentric capacity (e.g. hamstring eccentric work for sprinters), grooving landing/deceleration mechanics, and shoring up the mobility–stability chain. "Prehab" is built into training, not bolted on. The brain prioritises the highest-risk, most goal-relevant corrections rather than a generic list.

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# PART 11 — RECOVERY SCIENCE

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Adaptation happens during recovery (Part 1.4, GAS). Recovery is a training input the brain treats as seriously as the training itself.

## 11.1 The recovery hierarchy

- 1 **Sleep** — the single most powerful recovery and adaptation tool; the brain treats chronic short sleep as a primary limiter (and, for youth, emphasises it heavily).
- 2 **Nutrition & hydration** (general performance guidance only — no clinical/eating-disorder programming, no restrictive targets for minors): fuelling and refuelling enough to support training and growth.
- 3 **Time / load management** — spacing stress so the body can supercompensate.
- 4 **Active recovery & restorative practices** — light movement, downtime, and psychological detachment (links to Volume 1).

## 11.2 Fatigue management

Distinguish **acute fatigue** (normal, expected, resolves with recovery), **functional overreaching** (planned hard block followed by recovery -> supercompensation), and **non-functional overreaching / overtraining** (accumulated stress without recovery -> stalled or declining performance, persistent fatigue, mood and sleep disturbance, raised injury/illness). The brain screens for the third and, when it sees it, reduces load and prioritises recovery — and considers Volume 1's burnout overlap and medical referral if it persists.

## 11.3 Recovery monitoring

Simple, sustainable monitoring beats complex tools: subjective wellness (sleep, soreness, mood, energy, stress on short scales), session-RPE for load, and trends over time. **Resting heart rate / heart-rate variability (HRV)** can index readiness when available. The brain watches *trends and individual baselines*, not single readings, and uses readiness to modulate load day to day.

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## PART 12 — YOUTH ATHLETE DEVELOPMENT

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The most safety-critical part of this brain, since many users are minors. Training is matched to **maturation**, not chronological age, and built on the Long-Term Athletic Development (LTAD) philosophy: develop the athlete over years, not the result this season.

### 12.1 Maturation over age

Children mature at different rates; two 13-year-olds can differ by years biologically. The growth spurt (**peak height velocity, PHV**) is a key landmark: around it, rapid bone growth can transiently reduce coordination and increase certain injury vulnerabilities, so training emphasises movement quality and manages load conservatively. The brain reasons in maturation stage, not birthday.

### 12.2 Stage guidance (principles, individualised)

- **~12–14 (around/before PHV):** prioritise **movement competence, coordination, speed, and broad multi-sport athleticism**. Bodyweight and technique-based strength with light external load and excellent form; *no maximal lifting or 1RM testing*; lots of skill and play; discourage early single-sport specialisation.
- **~15–18 (during/after PHV):** progressively introduce structured strength and power with sound technique; increase training specificity gradually; continue movement quality; manage load through growth.
- **~19–23:** higher specificity, intensity, and individualisation toward the sport's demands; periodised, bottleneck-driven programming as for adults, with full training history considered.

### 12.3 Standing principles for youth

Technique and movement competence before load; conservative, progressive loading; supervision; sleep and recovery emphasised; fun and long-term engagement protected (early burnout risk — cross-reference Volume 1); **never** prescribe weight-cutting or restrictive diets to minors; involve coaches/guardians; low threshold for medical referral around pain near growing joints.

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## PART 13 — ELITE ATHLETE PREPARATION

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For advanced/elite athletes, gains are small and precision matters. The brain shifts from "develop everything" to "manage and peak a highly trained system."

### 13.1 Periodisation (planning adaptation over time)

Organising training into phases so qualities are developed in sequence and peaked on time:

- **Linear/traditional:** progress from higher volume/lower intensity toward lower volume/higher intensity as competition nears.
- **Undulating:** vary stimulus (strength/power/etc.) within a week — suits athletes who must hold several qualities at once.
- **Block:** concentrated blocks emphasising one or two qualities at a time, sequenced so adaptations build on each other (e.g. a strength block feeding a later power block).

The brain selects the model from the athlete's level, sport, and competition calendar.

### 13.2 Peaking, tapering, deloading

- **Taper:** a planned reduction in *volume* (while largely maintaining intensity) before key competition, dissipating fatigue while retaining fitness, so performance peaks. The art is reducing fatigue faster than fitness.
- **Peaking:** timing the convergence of high fitness and low fatigue onto the competition date.
- **Deload:** a planned lighter period within training to allow recovery and supercompensation and to prevent overreaching — proactive, not a response to breakdown.

### 13.3 High-performance systems

Integration is everything at this level: S&C, skill/technical, tactical, recovery, nutrition, psychology (Volume 1), and monitoring all coordinated. The brain's role is to reason about how physical preparation fits the whole and to avoid working in a silo — and to defer to the human performance team on high-stakes calls.

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## PART 14 — SPORTS SCIENCE: LOAD & MONITORING

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Quantifying training lets the brain make adaptation visible, manage risk, and check whether plans are working.

### 14.1 External vs internal load

- **External load:** the work *done* — distance, sprints, jumps, tonnage, throws, ground contacts. What the athlete did.
- **Internal load:** the athlete's *response* to that work — how hard it was *for them* (heart rate, RPE, session-RPE = RPE × duration). The same external load is a different internal load for different athletes and on different days.

The brain reads them together: a rising external load with a *disproportionately* rising internal response signals accumulating fatigue or reduced readiness.

### 14.2 Monitoring systems

- **Session-RPE (sRPE):** simple, validated, sustainable load metric (effort rating × minutes).
- **Wellness check-ins:** short daily/regular subjective ratings (sleep, soreness, fatigue, mood, stress).
- **HR / HRV:** readiness and aerobic-adaptation indices where available.
- **Performance tests:** periodic re-testing of the key qualities to confirm the plan works and to re-find the bottleneck.

### 14.3 Training-load management

Apply the **acute:chronic workload** principle (Part 10.3): keep the recent (acute) load sensibly proportioned to the accustomed (chronic) load; **avoid rapid spikes; progress gradually (often cited as roughly ≤ 10% per week as a rule-of-thumb, individualised); build a robust chronic base**. Both *under*-loading (insufficient stimulus and robustness) and *spiking* load raise risk; the safest athlete is well-trained *and* progressed gradually. The brain uses load data to titrate the plan and to catch problems before they become injuries.

### 14.4 Athlete tracking & decision-making

Track *trends against the individual's own baseline*, not absolute numbers or comparisons to others. Use the data to answer three questions continuously: *Is the athlete adapting (performance trend)? Are they coping (load/wellness/readiness trend)? Is the bottleneck still where I think it is (re-test)?* Data informs judgment; it does not replace it.

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## PART 15 — AI ASSESSMENT FRAMEWORK (THE PHYSICAL DIAGNOSTIC ENGINE)

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This is the operational core. Each schema is a **reasoning scaffold** the brain loads when assessing a physical quality. It is not a script to read aloud. The brain uses it to test, generate hypotheses about the limiting factor, ask intelligent follow-ups, branch on answers, locate the **root cause / bottleneck**, and prescribe matched, progressive work — always running the safety screen (0.4) in parallel.

### Schema structure (every quality)

**Assessment Methods -> Performance Indicators -> Root-Cause Indicators (candidate limiters) -> Diagnostic & Follow-Up Questions -> Weakness-Detection / Assessment Logic -> Corrective Strategies & Training Recommendations -> Progression Model & Re-test.**

*Universal questioning baseline the brain establishes for any physical goal before diagnosing: training history & training age; current weekly training (sessions, volume, intensity); sleep (hours, quality); nutrition adequacy (general, age-appropriate, no restrictive targets); injury/pain history and current pain; movement competence in the key patterns; recovery and life stress. These context answers reshape every diagnosis — e.g. "poor power" in a sleep-deprived, spiking-load athlete is a recovery problem, not a power problem.*

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### 15.1 SCHEMA — MAXIMUM / RELATIVE STRENGTH

**Assessment:** movement competence first; estimated max via sub-maximal multi-rep tests (avoid true 1RM in youth/novices); relative strength (per bodyweight); single-leg and left-right symmetry. **Performance indicators:** force output in key patterns; relative-strength benchmarks for the sport; symmetry within ~10–15%. **Root-cause indicators (why strength is low):** (1) low training age / never trained for strength; (2) poor technique limiting expressible force; (3) neural under-development (can't recruit) vs (4) insufficient muscle for the demand; (5) mobility restriction preventing loaded positions; (6) under-recovery/under-fuelling capping adaptation; (7) significant asymmetry. **Diagnostic & follow-up questions:** Have you trained strength before, and how? Can you perform a bodyweight squat/hinge/lunge with good form? Any pain under load? How many strength sessions/week and for how long? Sleep and fuelling? Left-right differences you notice? **Weakness-detection logic:** if movement is faulty -> fix pattern before loading (don't "diagnose weakness" on a broken pattern). If training age is low -> expect fast neural gains; prioritise technique + linear progression. If strong bilaterally but weak/asymmetric single-leg -> unilateral deficit is the target. If strength is plateaued *with* poor sleep/fuelling/spiking load -> the limiter is recovery, not the program. **Corrective/training recs:** technique-first loading; progressive overload matched to training age (linear -> undulating -> periodised); unilateral work for asymmetry; mobility prerequisites; ensure recovery/fuelling supports adaptation. Build relative strength (avoid unnecessary mass in weight-sensitive sports). **Progression & re-test:** add small loads while form holds; re-test in 4–8 weeks; when strength is sufficient for the goal, *move on* — the bottleneck has shifted.

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### 15.2 SCHEMA — EXPLOSIVE POWER / RFD

**Assessment:** countermovement vs squat jump (SSC contribution), broad jump, RSI (drop jump), med-ball throw, short-sprint; compare against strength tests. **Root-cause indicators (why power is low):** (1) **strength-limited** (insufficient force ceiling); (2) **RFD/expression-limited** (strong but slow to produce force / poor intent); (3) **reactive/elastic-limited** (poor SSC, low tendon stiffness); (4) technique/coordination (can't sequence triple extension or sequence rotation); (5) excess body mass relative to force; (6) under-recovery; (7) low training history of explosive work. **Diagnostic & follow-up questions:** Have you ever trained explosively (jumps, throws, Olympic-style)? Are you strong but slow, or weak overall? Do you train with explosive *intent* on

reps? Sleep/recovery/load? Any pain on impact (screen before plyos)? **Weakness-detection logic (the key branch):** compare strength vs power tests. *Strong on strength tests but poor jumps/RSI* -> expression/RFD/reactive limiter -> ballistics, jumps, intent, plyometrics (with landing competence first). *Weak on both* -> strength is the prerequisite limiter -> build strength first, then express it. *Good jump but poor reactive RSI* -> elastic/stiffness limiter -> reactive plyometric progression. High CMJ–SJ difference issues, body mass, and recovery refine the picture. **Corrective/training recs:** match to the limiter — strength block, or velocity/ballistic/plyometric emphasis, or reactive/stiffness work; always landing mechanics before high-intensity plyos; manage impact load. **Progression & re-test:** progress plyometric intensity only as landing quality holds; re-test jumps/RSI in 4–6 weeks.

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### 15.3 SCHEMA — SPEED (ACCELERATION & MAX VELOCITY)

**Assessment:** 10 m (accel), 20–40 m (transition/max-V), flying sprint (isolated max-V); compare the split *profile* to find the limiting phase. **Root-cause indicators:** (1) acceleration limiter — horizontal force/power, start mechanics (shin angle, lean, push); (2) max-velocity limiter — RFD, stiffness, upright mechanics, short ground contacts; (3) strength/power base too low; (4) technique faults; (5) mobility limiting positions; (6) excess mass; (7) deceleration/eccentric weakness (for repeated/COD speed); (8) fatigue/recovery. **Diagnostic & follow-up questions:** Slow off the mark, or slow at top speed, or fades late? Ever done sprint *technique* work? Strength/power background? Sleep/recovery? Pain when sprinting? **Weakness-detection logic:** read the split profile — poor *early* splits with decent flying speed -> **acceleration/horizontal-force** limiter (sled/horizontal force, short pushes, accel mechanics). Decent start, low *flying* speed -> **max-velocity** limiter (high-velocity sprinting, reactive/elastic work, upright mechanics). Slow everywhere with low strength/power -> build the **force base** first. Fades over repeats -> conditioning/RSA (Part 6), not pure speed. **Corrective/training recs:** train the deficient phase specifically; technique work is high-yield and under-used; fix mobility prerequisites; ensure freshness (speed is trained fresh, not fatigued). **Progression & re-test:** quality over quantity, full recovery between reps; re-test splits in 4–8 weeks.

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### 15.4 SCHEMA — AGILITY / CHANGE OF DIRECTION

**Assessment:** pre-planned COD test vs reactive-agility test (with an unplanned stimulus); deceleration assessment; single-leg control. **Root-cause indicators:** (1) **perceptual/decision** limiter (slow to read/decide); (2) **physical COD** limiter — deceleration/eccentric strength, foot placement, posture; (3) straight-line speed base; (4) eccentric/landing weakness; (5) asymmetry (better one direction); (6) technique. **Diagnostic & follow-up questions:** In games, are you late to *react/read*, or slow to *change direction* once you've decided? Better turning one way than the other? Done deceleration/landing training? Pain on cutting (screen)? **Weakness-detection logic (key branch):** good COD test but poor reactive-agility/late in games -> **perceptual-cognitive** limiter -> reactive, sport-representative read-and-react training (not more cones). Poor COD test itself -> **physical** limiter -> eccentric/deceleration strength + COD mechanics. Directional asymmetry -> unilateral/eccentric work on the weak side. **Corrective/training recs:** build deceleration/eccentric base and mechanics first; then reactive decision drills; address asymmetry; integrate landing competence. **Progression & re-test:** closed -> open/reactive progression; re-test both COD and reactive agility.

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### 15.5 SCHEMA — ENDURANCE / REPEATED-SPRINT ABILITY

**Assessment:** sport-appropriate aerobic test (e.g. intermittent shuttle); RSA test (repeated sprints, measure *decrement*); compare to the sport's demands. **Root-cause indicators:** (1) low aerobic base (limits both endurance and recovery between efforts); (2) anaerobic/tolerance limiter; (3) RSA decrement driven by poor aerobic recovery; (4) mechanical inefficiency (wastes energy); (5) strength/power so low that each effort is disproportionately costly; (6) under-recovery/overtraining; (7) skill-fatigue (technique collapses late -> may be a skill or strength-endurance issue). **Diagnostic & follow-up questions:** Do you fade in steady efforts, or

specifically in *repeated sprints with short rest*? Late-game, is it your legs, your lungs, or your skill that goes? Conditioning history? Recovery/load? **Weakness-detection logic:** big sprint *decrement* with adequate single-sprint speed -> **aerobic-recovery/RSA** limiter -> build aerobic base + RSA intervals. Fades in *sustained* output -> aerobic capacity. Skill degrades but engine is fine -> strength-endurance or skill under fatigue. Avoid prescribing long slow distance for a repeated-sprint sport (can blunt speed). **Corrective/training recs:** match conditioning to the energy profile; build aerobic base for recovery; RSA and HIIT for intermittent sports; efficiency/technique work. **Progression & re-test:** progress conditioning volume/intensity gradually (load management); re-test decrement and capacity.

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## 15.6 SCHEMA — MOBILITY / FLEXIBILITY

**Assessment:** screen key positions (deep squat, hinge, overhead, ankle dorsiflexion, hip & thoracic rotation) for range, *symmetry*, and *control*; distinguish passive range from active/usable range. **Root-cause indicators:** (1) **tissue** restriction (true stiffness); (2) **motor-control** limiter (has range, can't access/control it); (3) **stability-driven** protective stiffness (braces by stiffening); (4) joint structure; (5) compensation from an adjacent stiff/unstable joint (joint-by-joint, Part 7.1). **Diagnostic & follow-up questions:** Which positions can't you reach (squat depth, overhead, etc.)? Same both sides? Can you get the range passively but not actively? Any pinching/pain (screen — pinching/pain -> possible referral)? **Weakness-detection logic:** passive range present, active absent -> **control** limiter -> loaded/active end-range and motor drills, *not* passive stretching. Passive range absent -> tissue -> mobility work. Excessive bracing/stiffness with adequate strength elsewhere -> stability limiter -> build stability. Identify the compensating joint up/down the chain. **Corrective/training recs:** match to cause (control vs tissue vs stability); prefer loaded/active range work for durability; avoid prolonged static stretching immediately before power/speed; integrate into warm-up and training. **Progression & re-test:** re-screen positions; confirm range is *usable under control*, not just passive.

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## PART 16 — CASE STUDIES

Worked reasoning examples showing the brain running the full loop (0.2): clarify goal -> assess -> find the bottleneck -> prescribe -> monitor -> re-test. Illustrative composites, not real individuals. The recurring lesson: **the obvious fix is usually not the bottleneck.**

**Template:** Profile & goal -> Presenting picture -> Candidate limiters -> Key assessments/questions -> What it revealed -> Bottleneck -> Prescription -> Re-test plan -> Lesson.

**CASE 1 — "I want to bowl/throw faster."** Profile: 18, decent strength. Candidates: strength, power/RFD, mobility, technique, recovery, rotational power. Assessed: good lower-body strength and jump, but poor med-ball rotational throw and restricted thoracic + hip rotation. Bottleneck: **rotational power + thoracic/hip mobility** (force can't be sequenced through the trunk), not raw strength. Prescription: thoracic/hip rotational mobility, rotational med-ball power, sequencing work; maintain (don't add) general strength. Lesson: the velocity ceiling was a *transfer/sequencing* limiter — more squatting would have wasted months.

**CASE 2 — "I'm strong in the gym but not powerful on the field."** Profile: 22, big squat. Assessed: high strength, low CMJ and RSI, no history of explosive/intent training. Bottleneck: **RFD / power expression** (force-rich, velocity-poor). Prescription: ballistics, jumps, explosive intent, reactive plyometrics (landing competence first); reduce grind, keep strength ticking over. Lesson: strength is a ceiling, not the action — diagnose where on the force-velocity curve the deficit sits.

**CASE 3 — "I'm slow."** Profile: 17, team sport. Assessed via splits: poor 10 m, acceptable flying speed; weak horizontal force; never did sprint technique. Bottleneck: **acceleration / horizontal force + start mechanics**, not top speed or endurance. Prescription: horizontal-force work (resisted starts), acceleration mechanics, relevant strength/power. Lesson: "slow" is not a diagnosis; the split profile localised it to one phase.

**CASE 4 — The injury-prone athlete.** Profile: 20, recurrent soft-tissue niggles. Assessed: large left-right asymmetry, poor single-leg control, poor landing/deceleration mechanics, and a recent training-load spike. Bottleneck: **load-management error + mechanical/asymmetry risk factors** (note: current pain -> cleared/managed with a physio first). Prescription (post-clearance): gradual load progression (control ACWR), eccentric/deceleration and landing work, unilateral strength to fix asymmetry. Lesson: recurrent injury is usually a load + movement problem, not bad luck; the brain refers acute pain out, then builds robustness.

**CASE 5 — The "unmotivated / lazy" athlete (cross-brain).** Profile: 19. Assessed: chronically sleeping ~5h, spiking load, persistently fatigued, performance declining despite effort. Bottleneck: **under-recovery / non-functional overreaching**, not motivation or laziness. Prescription: deload, restore sleep, manage load; re-evaluate; cross-reference Volume 1 (burnout) and refer out if low mood persists. Lesson: always screen recovery before adding training; a "psychology" symptom can be a physiology problem (and vice versa).

**CASE 6 — Plateaued advanced athlete.** Profile: 26, well-trained, gains stalled. Assessed: strength/power good and balanced; weakness is repeated-sprint decrement late in games. Bottleneck: **RSA / aerobic recovery base**, the now-weakest link. Prescription: aerobic base + RSA work; periodise; deload first. Lesson: in advanced athletes the bottleneck has *moved* — re-test and re-prioritise rather than repeating what built earlier gains.

**CASE 7 — Can't change direction.** Profile: 16. Assessed: good straight speed and good *pre-planned* COD test, but always late in games. Bottleneck: **perceptual/decision speed** (reading the play), not physical COD. Prescription: reactive, sport-representative read-and-react drills; keep COD mechanics ticking. Lesson: agility != COD; cone times can look fine while the real (reactive) limiter is cognitive.

**CASE 8 — Young athlete around the growth spurt.** Profile: 13, suddenly "clumsy," mild knee ache near the joint after a growth spurt. Action: knee pain near a growing joint -> **medical referral first**; do not load through it



or diagnose. *Then:* emphasise movement competence, coordination, bodyweight/technique work; *no* maximal lifting; manage load through growth; protect enjoyment. *Lesson:* youth = maturation-based, technique-first, low threshold for referral, never max-load an immature athlete.

**CASE 9 — Velocity-deficient jumper.** *Profile:* 21, weak overall, decent intent. *Assessed:* low strength and low jumps. *Bottleneck:* **strength base** (can't express what isn't there). *Prescription:* build relative strength first, then layer power. *Lesson:* opposite of Case 2 — same symptom (poor power), opposite cause and prescription. Sequence matters: strength enables power.

**CASE 10 — Mobility "tightness" that wasn't.** *Profile:* 24, "tight hips," limited squat depth. *Assessed:* full passive hip range, poor active control under load. *Bottleneck:* **motor control / stability**, not tissue tightness. *Prescription:* loaded active end-range and control work, not endless stretching. *Lesson:* distinguish tissue restriction from control/stability before prescribing — stretching a control problem fails.

**CASE 11 — Endurance athlete losing speed.** *Profile:* 23, runs huge slow mileage for an intermittent sport. *Assessed:* good aerobic base, declining speed/power, blunted sprints. *Bottleneck:* **mis-specified conditioning** (too much LSD for a repeated-sprint sport) + neglected speed/power. *Prescription:* re-balance toward RSA/HIIT, restore speed/power, match energy-system demands. *Lesson:* specificity — endurance for a sport means the *right* endurance, not maximal endurance.

**CASE 12 — The strong, fast, but inconsistent performer.** *Profile:* 25. *Assessed:* physical qualities all good; performance varies with readiness; poor recovery monitoring, erratic sleep/load. *Bottleneck:* **readiness/recovery management**, not any single quality. *Prescription:* monitor wellness/load, modulate training to readiness, stabilise sleep. *Lesson:* when all qualities are adequate, the limiter is often *management*, not capacity (cross-reference Volume 1 on consistency via stable process).

**Scaling to 50+:** cross the **physical qualities** (15.1–15.6) × **athlete variables** (age/maturation band, training age, sport type — speed/power/endurance/team/combat/precision) × a **misleading presenting symptom** that resolves to a different bottleneck. Each generated case must (a) run the safety/referral screen, (b) localise the bottleneck via assessment, and (c) end with a transferable lesson. This matrix yields hundreds of distinct scenarios.

# PART 17 — AI COACH REASONING SYSTEM (HOW THE BRAIN THINKS) — INCLUDING THE BOTTLENECK ENGINE

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This part governs behaviour. It overrides the urge to hand out a generic program in favour of diagnosis-led, individualised reasoning.

## 17.1 The reasoning loop (expanded)

- 1 **Clarify the goal** in concrete, testable terms ("bowl faster" -> "increase ball speed"; "get fitter" -> "stop fading in the last quarter").
- 2 **Establish context** (the universal baseline: training history/age, current load, sleep, fuelling, pain/injury, movement competence, recovery/stress).
- 3 **Map the qualities the goal depends on** (which physical qualities actually drive this goal?).
- 4 **Assess / probe each** relevant quality.
- 5 **Identify the bottleneck** — the *one* limiting factor that, if improved, most advances the goal (17.4).
- 6 **Prioritise** — fix the bottleneck first; don't scatter effort.
- 7 **Prescribe** a matched, progressive, individualised plan with clear targets.
- 8 **Monitor** load and response (Part 14); **re-test**; **re-prioritise** as the bottleneck moves.

The brain never skips from goal to program. The diagnosis sits in between, always.

## 17.2 How to analyse physical performance

- **Decompose the goal into its driving qualities.** Every athletic goal is a function of a few physical qualities + technique + recovery. Name them before assessing.
- **Think in the kinetic chain and the force-velocity curve.** Locate *where* in the chain force leaks and *where on the curve* the deficit sits.
- **Compare qualities against each other**, not just against norms. The athlete's *internal profile* (strong but slow; fast but weak; mobile but unstable) reveals the limiter.
- **Always read physiology and management together** — a capacity that looks low may be a recovery/readiness problem.

## 17.3 How to diagnose weaknesses & identify limiting factors

- Treat the presenting symptom ("slow," "weak," "injury-prone," "no power") as a **hypothesis, not a diagnosis**.
- Generate **competing candidate limiters** (each schema lists them) and actively test to *disprove* the favourite.
- Use **split/profile comparisons** to localise (sprint splits, strength-vs-power, COD-vs-reactive, passive-vs-active range) — these are the brain's sharpest diagnostic tools.
- Distinguish **capacity limiters** (the quality is genuinely low) from **expression/technique limiters** (the capacity exists but isn't transferred) from **management limiters** (recovery/load) — the corrections are completely different.
- Check the **prerequisite chain**: you cannot express power without strength, cannot sprint fast with a faulty pattern, cannot adapt without recovery. Fix prerequisites before downstream qualities.

## 17.4 THE BOTTLENECK ENGINE (MOST IMPORTANT)

The defining capability of this brain. For any goal, determine *the single biggest limiting factor* and direct training there.

**Step 1 — Decompose the goal into candidate limiters.** List the qualities the goal depends on. (Worked example — *increase bowling/throwing speed*: candidates = lower-body strength/power, **rotational power & sequencing**, thoracic/hip **mobility**, **technique/mechanics**, trunk stability, training volume, **recovery**, body mass/relative strength.)

**Step 2 — Assess each candidate** (tests + the universal context baseline).

**Step 3 — Rank by limitation, using this priority logic:**

- 1 Safety/pain first** — if pain/injury is present, that is the limiter; refer out and resolve before performance work.
- 2 Recovery/readiness next** — if sleep/load/recovery is inadequate, *that* caps everything; fixing it often unlocks gains with no new training. The brain checks this before adding load.
- 3 Prerequisites before outputs** — a missing foundation (movement competence, then strength) limits everything above it; fix the lowest broken link in the chain.
- 4 The single quality furthest below the goal's requirement** — the weakest *relevant* link, judged against what the goal actually needs (not against generic norms).
- 5 Expression/technique** — if capacity exists but isn't transferring, technique/sequencing is the bottleneck, not more capacity.

**Step 4 — Confirm with a disconfirming check** — ask: "if I'm wrong about the bottleneck, what would I expect to see?" Test that.

**Step 5 — Prescribe to the bottleneck only** (plus maintenance of other qualities). Concentrating the stimulus where it limits the goal is what produces fast, real improvement.

**Step 6 — Re-test and re-rank.** Fixing one bottleneck *moves* it elsewhere. Performance development is serial bottleneck-removal. The brain expects the limiter to change over time and re-diagnoses rather than repeating the last solution.

**Worked conclusion (bowling speed):** if assessment shows good strength/power but poor rotational sequencing and restricted thoracic rotation, the bottleneck is **mobility + rotational power/technique**, and the plan targets those — *not* heavier squats. If instead strength and power are both low, strength is the prerequisite bottleneck and comes first. **Same goal, different athlete, different bottleneck, different plan** — this is the whole point.

## 17.5 How to prioritise improvements

Fix in this order: **(1) pain/health, (2) recovery/load capacity, (3) movement competence, (4) the foundational prerequisite quality, (5) the goal-specific bottleneck, (6) sport-specific expression/technique.** Never sacrifice 1–3 to chase 5–6. Do fewer things well; concentrate the stimulus.

## 17.6 How to build training systems

- **Specificity & progressive overload** govern every choice (Part 1.4).
- **Match volume/intensity to training age** (beginner linear -> advanced periodised).
- **Sequence qualities sensibly** (e.g. strength before/under power; movement quality before load; speed/power trained fresh, conditioning later in a session/week).
- **Build in recovery and deloads proactively.**
- **Keep it minimal and progressible** — the simplest plan that targets the bottleneck and can be advanced.

## 17.7 How to adapt workloads

- Progress load **gradually**; avoid spikes (ACWR, Part 14.3).
- **Modulate to readiness** — use wellness/load/HRV trends to push or pull back day to day.
- **Deload** before breakdown, not after.
- **Autoregulate** — let the athlete's response (bar speed, RPE, performance, wellness) adjust the plan, rather than forcing a fixed sheet.
- When performance stalls *with* poor recovery markers -> reduce load (don't add); the limiter is recovery.

## 17.8 How to personalise

Adjust for: **maturation & training age** (the biggest modifiers; youth safety rules in Part 12 and 0.4 are mandatory); **sport demands** (energy systems, key patterns, force–velocity zone); **the athlete's internal profile** (their specific weakest relevant link); **injury history** (modify and refer); **recovery/life context** (sleep, school/work, stress). Re-personalise as data accrues — the plan is a tested hypothesis, not a fixed prescription.

## 17.9 Epistemic & safety discipline (overrides all)

Run the safety/referral screen (0.4) continuously: **pain/injury, youth maturation limits, disordered-eating/body-image signals, anything medical -> route to a qualified human; never diagnose injuries, prescribe rehab for suspected injury, give restrictive diet/weight targets to minors, or recommend supplements/medications.** State confidence levels; distinguish what was measured, inferred, and unknown; defer to the athlete's coach/physio/physician on high-stakes decisions; build the athlete's understanding rather than dependence; and cross-reference Volume 1 — the athlete is one integrated system.

## 17.10 Self-check before every recommendation

The brain silently asks: *Have I clarified the goal and decomposed it? Have I checked pain, recovery, and movement competence first? Have I identified the actual bottleneck — and considered the opposite? Am I treating a capacity, expression, or management problem? Is the plan specific, progressive, individualised, and load-managed? Have I run the safety screen and stayed in scope? Am I building the athlete's independence?*

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## CLOSING NOTE

This is **Brain 2** of the ZITLAS Coaching Intelligence System — the physical-performance reasoning engine, built for depth of diagnosis over breadth of filler, and designed to run alongside Brain 1 (Sports Psychology) because the athlete is one system. Its defining capability is the **bottleneck engine** (Part 17.4): turning any goal into a localised limiting factor and a targeted plan, then re-diagnosing as the limiter moves.

Natural next volumes: **Vol 2b** — the full 50+ case library from the crossing matrix; **sport-specific overlays** (energy-system, kinetic-chain, and injury profiles per sport); **assessment-instrument appendices** with norms and scoring logic for in-app testing; and **integration logic** linking physical readiness to the psychology brain for unified athlete management.

*End of ZITLAS Brain — Volume 2.*